KARUNA VASU

BACKEND/PLATFORM ENGINEER | PYTHON | AUTOMATION SYSTEMS | AWS | AI SYSTEMS

CONTACT

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WWW: Bold Profile

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WWW: https://karunaportfolio-tau.vercel.app/

SKILLS

Languages: Python, SQL, JavaScript, TypeScript, C#, PERL, R
Backend & APIs: FastAPI, REST APIs, microservices, ASP.NET Core, Flask, Django, SQLAlchemy
Frontend: React, Next.js, Vite, Tailwind CSS
Automation: Selenium, RPA, workflow automation
AI/ML: OpenAI APIs, LLMs, prompt engineering, document AI, Amazon Bedrock (Claude 3.5 Sonnet)
Data Processing: Pandas, NumPy, rule engines, data validation
Cloud & DevOps: AWS (Connect, Bedrock, Lambda, EC2, DynamoDB, IAM), Docker, CI/CD, Azure Key Vault
Databases: PostgreSQL, Microsoft SQL Server, SQL, SQLite, vector DB (basic)
Platform & Tooling: APScheduler, Hangfire, Serilog / structured logging, PyODBC, SFTP automation, Git, Linux, Postman

EDUCATION

January 2023
Master of Science: Bioinformatics
Manipal Academy of Higher Education, Manipal, India

PROFESSIONAL SUMMARY

Backend-focused Python Developer with 2+ years of experience building scalable automation platforms, AI-driven workflows, and microservices-based systems in US Healthcare (RCM).
Proven impact:

- - Reduced processing time by up to ~95% (e.g., Availity-integrated claim pipelines)
- - Improved operational efficiency by ~40% on key workflows (EHR, platform reliability)
- - Automated high-volume workflows, reducing manual effort by ~40-80% (including IVR calling).

Specialized in FastAPI microservices, distributed bot orchestration, AI-powered document processing (AWS Bedrock), voice/IVR automation (Amazon Connect), and real-time workflow automation. Contributed across BotVeta's orchestration API, operator and admin frontends, and integrated Python services.

EXPERIENCE

December 2023 - Current
Python Developer, ***Integrity Healthcare Solutions Pvt Ltd***, Ahmedabad

- **Healthcare RCM & Automation**
- -Architected and delivered scalable automation for US Healthcare RCM (eligibility,authorization,claims), reducing manual operations through high-volume pipeline design.
- - Delivered and maintained automation across 156+ Azure DevOps repositories in the Automation project covering payer/client workflows.
- -Automated processes including eligibility, authorization, EOB/ERA extraction, charge/payment posting, report downloading, note pasting, and rule-engine input creation.
- - Built high-performance automation pipelines integrated with Availity, accelerating claim processing turnaround by up to ~95%.
- -Designed distributed bot orchestration with queue management, retry logic, centralized logging, and fault-tolerant execution.
- -Developed production-grade microservices using FastAPI for scalable, low-latency backend processing; improved system performance and reliability on key workflows by ~40%.
- - Automated EHR workflows (Practice Fusion), streamlining operations and improving efficiency by ~40%.
- - Implemented reusable automation frameworks and validation pipelines, reducing regression effort by ~80%.

BotVeta (Bot Manager / orchestration platform)

- - Contributed to BotVeta - centralized scheduling, monitoring, and scaling of enterprise automation workflows across organizations.

- Electrocardiogram Heart Rate Variability Signal Analysis and Data Acquisition - Research Project
- GPA: 7.24

January 2021
Bachelor of Science: Bioinformatics
Sardar Patel University Gujarat, Anand, India
GPA: 7.00

GITHUB

Other:
https://github.com/KarunaV23

CERTIFICATIONS

Jira Fundamentals Badge, Atlassian

Jira Fundamentals Assessment, Atlassian

KPMG_Data Analytics Consulting Virtual Internship, Forage

Deep Learning, Manipal Institute of Technology

Skills for Healthcare Data Analysts, LinkedIn

Linux System Engineer: Advanced Disk Systems and System Backup, LinkedIn

SAS Essential Training

Descriptive Analysis for Healthcare Research, LinkedIn

The Non-Technical Skills of Effective Data Scientists, LinkedIn

Tableau Essential Training, LinkedIn

Machine Learning for Intelligent Systems, IEEE Robotics and Automation Society

Google Foundations: Data, Data, everywhere, Coursera

Drug Discovery, Coursera

JavaScript Basics, Coursera

Clinical Trials Data Management and Quality Assurance, Coursera

PUBLICATIONS

- - Built and integrated bot-manager services (FastAPI + Next.js + TypeScript) with JWT RBAC, APScheduler-based timed runs (once/daily/weekly + timezone), and run/job lifecycle tracking.
- - Supported mixed execution backends (Python Flask and Node.js bot servers) with trigger/stop APIs, PID-based control, and health/status routes for reliable runtime operations.
- - Designed and developed scalable bot orchestration systems with scheduling, lifecycle management, and execution tracking.
- - Implemented fault-tolerant workflows with retry handling, checkpoint-based recovery, and centralized monitoring.
- - Unified Python and Node.js bot runtimes through API-driven orchestration for improved interoperability.
- - Integrated enterprise utilities including Azure Key Vault, AWS credential flows, SFTP, Gmail and SQL/ODBC pipelines.
- - Orchestration API (.NET 9): SQL Server, Entity Framework Core, JWT authentication, Hangfire scheduling, Serilog, Swagger/OpenAPI; multi-tenant APIs for bot library, instances, runs, logs, and workflows.
- - Operator UI: React 19, Vite, TypeScript - bot instances, run history, workflow authoring, user management; protected routing and REST integration.
- - Admin console: Next.js 16, React 19, TypeScript, Tailwind - organization setup, integrations, and PDF abstractor workflows.

- **AI PDF extraction & document intelligence (BotVeta Admin integration)**
- - Built AI-driven document processing integrated into BotVeta for healthcare RCM: admin-side upload (large PDFs, e.g. up to 200MB), patient- and document-type-aware grouping, split and bundle workflows, job progress tracking.
- - Full extraction via FastAPI (Python / uvicorn) with AWS Bedrock; analyze-only mode; rule-based + confidence-aware validation for splits/grouping; JSON and Excel outputs for EOB, claims, and patient-centric data.

- **AI voice / IVR automation (Amazon Connect)**
- - Designed and deployed a scalable IVR automation platform using Amazon Connect for payer calls (claims, eligibility, authorization).
- - Integrated AI-driven conversational capabilities with Amazon Bedrock (Claude 3.5 Sonnet).
- - Built Python backends with REST and AWS (EC2, DynamoDB, Lambda) for real-time call orchestration; auto-navigation of IVR menus, response capture, dynamic routing to agents (including Amazon Connect CCP-aligned monitoring and routing patterns).
- - Built React.js frontends for BotVeta IVR dashboards
- - real-time monitoring and control of call pipelines.
- - Excel-driven workflow triggers for non-technical users; reduced manual calling effort by ~60-80% and improved response turnaround; IAM-aligned secure cloud deployment.

June 2023 - July 2023
Data Science Intern, ***Zephyr Technologies and Solutions Pvt Ltd***, Mangalore, India

- Developed real-time chat application using Django, WebSockets, and PostgreSQL.

- Electrocardiogram Heart Rate Variability Signal Analysis and Data Acquisition: A Novel Approach for Sympathovagal Inference Vasu, K., Kumar, P., Mane, P. R., Paul, B., Vaishali, K., Sinha, M. K. 2023 International Conference on Network, Multimedia and Information Technology (NMITCON), 09/01/2023.
- DOI: https://doi.org/10.1109/NMITCON58196.2023.10276059
- Best Paper Award, International Conference on Network, Multimedia and Information Technology (NMITCON), 2023-09-01, Outstanding contribution to the field.
- Best Presenter, International Conference on Network, Multimedia and Information Technology (NMITCON), 2023-09-01

KEY ACHIEVEMENTS

- Delivered end-to-end automation systems processing high-volume healthcare workflows with up to 95% reduction in turnaround time.
- Designed and implemented a multi-tenant orchestration platform (BotVeta) supporting scalable workflow execution and monitoring.
- Built and deployed AI-powered production systems using AWS Bedrock for document intelligence and IVR automation.
- Developed distributed, fault-tolerant backend systems with queue-based execution, retry handling, and real-time orchestration

February 2023 - December 2023
Research Intern, ***Manipal Institute of Technology***, Manipal, India

- - Published IEEE research paper on ECG signal analysis (see Publications).
- - Awarded Best Paper & Best Presenter (NMITCON 2023).
- https://doi.org/10.1109/NMITCON58196.2023.10276059

KEY PROJECTS

BotVeta - Automation Orchestration Platform (Flagship)

- - Multi-tenant platform for scheduling, monitoring, and managing distributed automation workflows.
- - Architecture: Orchestration APIs -> Scheduler -> Bot Runners -> Centralized logging & monitoring.
- - Supports queue-based execution, retry handling, failure recovery, and workflow orchestration.
- - Tech: Python, FastAPI, .NET, React, Next.js, SQL Server, PostgreSQL, AWS, Docker.

AI-Powered PDF Extraction & Document Intelligence System

- - AI-driven document processing system for extracting structured data from unstructured healthcare PDFs.
- - Implements document splitting, grouping, and validation using rule-based logic with AI confidence scoring.
- - Integrates AWS Bedrock (Claude) for LLM-based extraction.
- - Tech: Python, FastAPI, AWS Bedrock, LLMs.

AI Voice Automation Platform (IVR + Amazon Connect)

- - Real-time IVR automation system for payer workflows.
- - Integrates Amazon Connect with backend orchestration and AI-driven conversational flows.
- - Supports automated navigation, response handling, and dynamic call routing.
- - Tech: Python, REST APIs, Amazon Connect, AWS.

RCM Automation Suite (Eligibility - Authorization - Claims)

- - Automation platform for eligibility, authorization, and claims workflows.
- - Implements rule-based validation, retry mechanisms, and workflow orchestration.
- - 40-80% reduction in repetitive effort.
- - Tech: Python, Selenium, Pandas, SQL.

Slack-Based Workflow Automation System

- Event-driven system for triggering and monitoring workflows via Slack.
- Uses APIs and webhooks for real-time execution.
- Tech: Python, FastAPI, Slack API.

AI Bots & Utility Systems

- - Built conversational bots (Telegram/WhatsApp) using LLMs.
- - Developed audio transcription pipelines using OpenAI APIs.
- - Tech: Python, OpenAI APIs, Flask.